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BATTERY MODULES AND SYSTEMS FOR REMOTE COMMAND AND CONTROL OF SAME

Abstract

Described are remote command-enabled battery modules and systems and methods incorporating them.

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Background/Summary

[0001] This application is a continuation of Ser. No. 17/394,943, filed Aug. 5, 2021, which is a continuation of Ser. No. 16/220,142, filed Dec. 14, 2018, which is a continuation of PCT/2017/037495, filed Jun. 14, 2017 which claims the benefit of priority of U.S. Provisional Patent Application No. 62/349,761, filed Jun. 14, 2016, which are all hereby incorporated herein by reference in their entirety.

BACKGROUND

[0002] The present invention relates generally to batteries and uses thereof, and in certain embodiments to battery modules including unique features that are useful in achieving remote monitoring, command and control of the battery modules, for example by multiple remote users at dispersed locations.

[0003] Technologies in the fields of battery materials and chemistries have advanced significantly in the past few decades. The use of batteries, off-grid, to power a multitude of transportation and utility apparatuses is growing. Many of these uses, however, put high physical and performance demands upon the batteries. This can be particularly true in regions where grid power has traditionally been unavailable.

[0004] Batteries in general, and more particularly lithium-ion batteries, present challenges. For instance, variations in electrical storage with operating temperatures are often experienced which can decrease battery life and performance. Lithium-ion and other batteries can lose capacity at an accelerated rate when stored or operated in higher temperature environments. Complicating the matter is the fact that some lithium-ion batteries tend to increase temperature during operation due to Joule heating within the batteries. As the temperature of some lithium-ion batteries increases to a destructive level, cells within the batteries can become unstable and begin internally discharging across their negative and positive terminals. This discharge can generate Joule heating and warm the battery further. In turn, the increasing temperature causes further instability, discharge, and (potentially) the loss of the battery.

[0005] Another challenge with many modern battery designs stems from the fact that battery cell constructions are physically vulnerable. Illustratively, lithium ion battery chemistry is often encompassed in a “soft pouch” or “pouch cell” format, where the internal components and chemistries are incorporated within a flexible polymeric pouch. If this pouch is damaged, leaks can result which can potentially create electrical “short” circuits that can lead to battery loss. Similar

issues can be experienced with other mechanically vulnerable battery cell constructions.

[0006] Implementation and widespread adoption of battery power in communities has been slow in developing, which is possibly due at least in part to issues related to capital expense, technology complications and user impressions in respect of system reliability and flexibility in application.

[0007] In light of the background in this area, needs exist for improved and/or alternative battery modules enabled for remote monitoring, command and control, as well as systems implementing use of such battery modules. In some of its aspects, the present disclosure is addressed to these needs.

SUMMARY

[0008] In one aspect, provided is a battery module for remote command. The battery module includes a battery case and a plurality of battery cells received within the battery case and electrically connected to one another. The battery module also includes a negative electrical terminal electrically coupled to the battery cells and a positive electrical terminal electrically coupled to the battery cells. The battery cells provide a voltage potential between the positive electrical terminal and the negative electrical terminal. The battery module also includes electronics received within the battery case and configured to enable electromagnetic signaling to and from the battery module, and a disable mechanism responsive to the electromagnetic signaling to the battery module and operable to disable use of the voltage potential. In some forms, the disable mechanism interrupts electrical continuity between the positive electrical terminal and the negative electrical terminal. In other forms, the disable mechanism is operable to shield at least one of the positive electrical terminal and the negative electrical terminal from external electrical contact.

[0009] In another aspect, provided is a battery module for remote command. The battery module includes a battery case and a plurality of battery cells received within the battery case and electrically connected to one another. The battery module also includes a negative electrical terminal electrically coupled to the battery cells and a positive electrical terminal electrically coupled to the battery cells. The battery cells provide a voltage potential between the positive electrical terminal and the negative electrical terminal. The battery module also includes electronics received within the battery case and configured to enable electromagnetic signaling to and from the battery module. The electronics are also operable to generate a value representative of cumulative power drawn from the battery cells over a period of time and to transmit the value to a remote location by the electromagnetic signaling from the battery module. The battery module can also include a disable mechanism responsive to the electromagnetic signaling to the battery module and operable to disable use of a voltage potential between the positive electrical terminal and the negative electrical terminal. The battery module can also include a plurality of visible indicators, for example lights, mounted to the battery case for signaling a condition of the battery module to a user.

[0010] In another aspect, provided is a system for management and control of one or more battery modules. The system includes one or more battery modules including a battery case, a plurality of battery cells received within the battery case and electrically connected to one another. The battery module(s) also include a positive electrical terminal electrically coupled to the battery cells and a negative electrical terminal electrically coupled to the battery cells. The battery cells provide a voltage potential between the positive electrical terminal and negative electrical terminal. The battery module also includes electronics received within the battery case and configured to enable electromagnetic signaling to and from the battery module. The system further includes a controller remote from and communicatively coupled to the one or more battery modules, with the controller being configured to receive the electromagnetic signaling from the battery module and transmit the electromagnetic signaling to the battery module.

[0011] In aspects herein including or employing a battery module(s), the battery module(s) can include a polymeric case defining a plurality of openings in a wall thereof. A shock dampening material is received within the polymeric case, and a monolithic, thermally-conductive, reinforcing

divider is at least partially received within the polymeric case. The monolithic, thermally-conductive, reinforcing divider is arranged to structurally reinforce the polymeric case, with the shock dampening material positioned between the reinforcing divider and the polymeric case and operable to dampen and transfer forces between the polymeric case and the reinforcing divider. The monolithic, thermally-conductive, reinforcing divider includes a plurality of dividing wall members defining a plurality of cavities therebetween, with the cavities located within the polymeric case. The monolithic, thermally-conductive reinforcing divider also defines a plurality of thermal transfer projections received through respective ones of the openings of the polymeric case and providing thermal transfer members projecting externally of the polymeric case. Lithium ion battery pouch cells are received within the polymeric case and include lithium ion battery pouch cells received at least partially in respective ones of the divider cavities. An electronic control board is received within the polymeric case and electrically couples the lithium ion battery pouch cells in series to provide a battery pouch cell series or module. Negative and positive electrical terminals are electrically coupled to the battery pouch cell series on respective ends thereof. In preferred forms, the reinforcing divider is composed of extruded aluminum, which can potentially be modified post-extrusion, e.g. by machining, to incorporate additional features. The reinforcing divider can also define at least one external reinforcing scaffolding structure that includes a plurality of elongate scaffold walls extending in a first direction and a plurality of cross-scaffold walls extending in a second direction transverse to said first direction and connecting the elongate scaffold walls. The battery case can include at least first and second case portions which are sealed together to form an enclosure, for example at mating peripheral rims or lips provided on each portion.

[0012] In other aspects herein, the battery module(s) of or employed in systems herein can include a battery case and a monolithic divider element at least partially received within the battery case. The monolithic divider element includes a plurality of dividing wall members defining a plurality of cavities therebetween. A plurality of battery cells are received within the case and include battery cells received at least partially in respective ones of the divider cavities and electrically connected to provide a battery cell series. A negative electrical terminal is electrically coupled to a first end of the battery cell series, and a positive electrical terminal is electrically coupled to a second end of the battery cell series. In certain embodiments, the divider element is composed of extruded aluminum, and or defines at least one external reinforcing scaffolding structure including a plurality of elongate scaffold walls extending in a first direction and a plurality of cross-scaffold walls extending in a second direction transverse to the first direction and connecting the elongate scaffold walls. In addition or alternatively, the battery module can include a shock dampening material positioned between the divider element and the battery case. The monolithic divider element can be only partially received in the battery case, and can define one or more thermal transfer elements, desirably a plurality of thermal transfer elements, exposed externally of the battery case. The transfer elements can be in heat transfer relationship with the dividing walls, and the dividing walls can be in heat transfer relationship with the battery cells.

[0013] In still other aspects herein, the battery module(s) employed can include a battery case and a divider element at least partially received within the battery case. The divider element includes a plurality of dividing wall members defining a plurality of cavities therebetween. A plurality of battery cells is received within the polymeric case and includes battery cells received at least partially in respective ones of said cavities and electrically connected in a battery cell series. A negative electrical terminal is electrically coupled to a first end of the battery cell series. A positive electrical terminal electrically coupled to a second end of the battery cell series.

[0014] Beneficial variants of the embodiments disclosed above in this Summary include those containing unique structural and/or functional features as described for the embodiments in the Detailed Description below.

[0015] Additional embodiments, as well as features advantages of aspects of the invention, will be apparent to persons of ordinary skill in the relevant art from the descriptions herein.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0016] FIG. 1A provides a schematic diagram of a battery module enabled for remote monitoring, command and control and a communication and control system associated with the battery module.

[0017] FIG. 1B provides a flow diagram of some embodiments implementing remote monitoring, command and control of one or more battery modules.

[0018] FIG. 1C provides a perspective view of one embodiment of a battery module illustrating features that may be included in remote control-enabled battery modules herein and systems associated with them.

[0019] FIG. 2 provides a top view of the battery module of FIG. 1C.

[0020] FIG. 3 provides a bottom view of the battery module of FIG. 1C.

[0021] FIG. 4 provides a partial cut-away view of the battery module shown in FIG. 1C and illustrating internal components thereof.

[0022] FIG. 4A provides an enlarged cut-away view of a section of the peripheral flange of the battery case as shown on the left side of FIG. 4.

[0023] FIG. 5 provides a front view of the reinforcing divider of the battery module of FIGS. 1C to 4.

[0024] FIG. 6 provides a top view of the reinforcing divider shown in FIG. 5.

[0025] FIG. 7 provides a bottom view of the reinforcing divider shown in FIG. 5.

[0026] FIG. 8 provides a view of the inside of the battery case tub of the battery case of the battery module of FIGS. 1C to 4.

[0027] FIG. 9 provides a plan view of the inside of the battery case cap member of the battery module of FIGS. 1C to 4.

DETAILED DESCRIPTION

[0028] For the purposes of promoting an understanding of the principles of the invention, reference will now be made to certain embodiments thereof and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended, and alterations and modifications in the referenced embodiments, and further applications of the principles of the invention as illustrated herein are contemplated as would normally occur to one skilled in the art to which the invention relates. In the discussions below, a number of features of a battery module(s) and/or a system for command and control of a battery module(s) are disclosed. It will be understood that any one, some or all of such disclosed features can be combined with the general embodiments discussed in the Summary above or set forth in the Claims below to arrive at additional disclosed embodiments herein. Features disclosed herein are to be understood to be combinable with each other unless it is clearly stated or it is clear from context that they are not combinable.

[0029] As disclosed above, in certain aspects, the present disclosure relates to battery modules that are enabled for remote monitoring, command and control, and to systems that incorporate such battery modules. With reference now to FIG. 1A, shown is one embodiment of a battery module that is enabled for remote monitoring, command and control, and a communication and control system **200** associated with the battery module. System **200** includes a battery module **202**. Battery module **202** can optionally have the features described for battery modules in conjunction with FIGS. 1C to 9 herein. Thus, battery module **202** can have a battery case **204** enclosing a plurality of battery cells **206**. Battery module **202** can also have thermal management features **208**, for example thermally conductive elements with portions thereof extending outside case **204** as disclosed

herein. Other thermal management features, for example those that involve circulating liquid or other fluid coolants within the battery case **204**, for example to remove heat generated by the battery cells, can also be used. Battery module **202** includes a positive terminal **210** and a negative terminal **212**. Battery module **202** also includes electronics that participate in and enable the communication and monitoring, command and control features described herein. Thus, battery module **202** can include an electronic gyroscope **214** or a plurality of such gyroscopes, which can detect and give indication of the orientation of the battery module **202**, e.g. with respect to gravity. Battery module **202** can also include a gravitational accelerometer **216**, or a plurality of such accelerometers, which can detect and provide indication of accelerations experienced by the battery module **202**. Battery module **202** can also include a control board **218** which, as discussed elsewhere herein, may be electrically connected to the battery cells **206** of battery module **202** and can function in both detecting conditions of and controlling usage of such battery cells **206**. Battery module **202** also includes a satellite and/or cellular antenna **220** enabling communication by satellite, cellular, or other electromagnetic signal-based communication systems, or a plurality of such antennas. The antenna or antennas are preferably housed within battery case **204**. In conjunction therewith, battery module **202** can include a satellite and/or cellular and/or GPS modem and VMS interface card **222**. This interface card **222** can be communicatively coupled to the other electronics of the battery module **202**, for example the control board **218**, the accelerometer **216**, and the gyroscope **214**. In this manner, a two way datalink may be enabled by which these electronics can communicate with and be controlled remotely by electromagnetic signal-based communication systems. Battery module **202** can also include an externally visible display **224**, for example as described elsewhere herein. Battery module **202** can also include a communications connector **226** by which wired communications can be made with battery module **202** and in particular its electronic detection, control and signaling elements. Battery module **202** also includes an on/off switch **228** by which the remote signaling to the battery module **202** can be used to disable the battery module **202** from use to draw power from battery module **202**. The switch **228** can be a dedicated circuit component such as a relay or the like, or can be a software based switch for example which can be controlled by circuit board **218**. Additionally, other means of disabling use of battery module **202** can be used. For example, the battery module **202** can include components which can be activated to prevent access to the positive terminal **210** and/or the negative terminal **212** to prevent a user drawing power from the battery. For example, the terminal or terminals could retract into the battery case, e.g. as driven by a motor mounted in or on the battery case **204**, to prevent access to one or both of them, and/or the battery module **202** can include a covering component, e.g. driven by a motor, which can be activated to cover one or both of the positive terminal **210** or negative terminal **212**, to prevent a user drawing power from battery module **202**. These and other ways to prevent use of the battery module **202** are contemplated. System **200** also includes electromagnetic signal-based communication links (e.g. cellular or satellites) through satellites **230** and/or cell towers **232**, for example routed through the internet **234**, to a remote data acquisition and command center **236**. Center **236** can include one or more computers or other controllers **238** (e.g. computers or other microprocessors), along with one or more user inputs **240** (e.g. keyboards) into the controller(s) **238**.

[0030] In some embodiments, in addition to or as an alternative to external display **224**, battery module **202** can include one or more indicators, for example lights, to signal a user of the battery module. The indicator(s) can be remotely activated and/or deactivated by the remote data acquisition and command center **236**. For example, the battery module can include lights **242**, **244** and **246** that are visible to a user from locations external of the battery module **202**. Lights **242**, **244** and **246** can be of different color from one another (for example green, yellow and red), and/or can be enabled for operation in constant on, flashing, or off conditions. These lights can be powered by battery cell(s) of the battery module **202**. These lights can be remotely controlled from command center **236** to signal a user of module **202** with respect to a condition of the battery module **202**

(e.g. as in a warning) and/or with respect to an action that will be taken with respect to battery module **202** at a future time, as remotely controlled by command center **236** (which may sometimes be referred to as headquarters or “HQ”). Illustratively, lights **242**, **244** and **246** can be green, yellow and red, respectively. The following series of indicator conditions can then be adopted:

TABLE-US-00001 Green light 242 constant Battery operation condition acceptable; no warnings to on, lights 244 and 246 provide to user off Green light 242 flashing Battery operation condition acceptable; yellow-level lights 244 and 246 off warning condition will ensue at a future time unless some action is taken (e.g. by user) Yellow light 244 Yellow-level (intermediate level) warning condition (e.g. constant on, lights 242 indicative of a condition of the battery module and/or of an and 246 off account associated with the battery module) Yellow light 244 Yellow-level (intermediate level) warning condition; red- flashing, lights 242 and level (high level) warning condition will ensue unless some 246 off action is taken (e.g. by user) Red light 246 constant Red level (high level) warning condition (e.g. indicative of on, lights 242 and 244 a condition of the battery module and/or of an account off associated with the battery module) Red light 246 flashing, Highest level warning condition (e.g. indicative of an lights 242 and 244 off imminent shut down or disabling of the battery module)

[0031] These and other strategies for using one or more lights mounted on and/or in battery module **202** to signal the user can be used in systems herein, for example to signal the user with respect to notifications or warnings as to the battery module and/or an associated user account as discussed elsewhere herein.

[0032] The system **200** can also include an electrical charging source **250** (e.g. a solar-powered, wind-powered, or electrical grid-powered source) coupled or capable of coupling to the battery module **202** to electrically charge the battery module **202**. The charging source **250** can be coupled or capable of coupling to the battery module **202** through an inverter or other suitable electrical component **252**. Component **252** can also be enabled for remote control by command center **236** via electromagnetic signaling, for example including its own satellite and/or cellular antenna **254**. In one mode, component **252** can include a switch, relay or other component (e.g. a dedicated component or software-implemented) by which remote signaling from center **236** can enable and/or disable charging of battery module **202** by charging source **250**. This can, for example, be implemented by command center **236** in response to a condition of the battery module **202** and/or in response to a condition of a user account associated with battery module **202**. Electrical connection of the battery module **202** to the charging source **250** can be through charging leads selectively connectable to positive and negative electrical terminals **210** and **212** and/or through a different, dedicated charging port provided on battery module **202**.

[0033] As discussed above, battery module(s) **202** are enabled for self-sensing or detection of battery module conditions and for electromagnetic signaling to report these conditions remotely to command center **236**. In this regard, the detection and signaling from the battery module can in some forms be representative of one of, any combination of some of, or all of: [0034] a. a voltage value, for example of the battery module, of a battery cell of the battery module, and/or of a battery cell series of the battery module; [0035] b. a temperature value, for example of the battery module, a region of the battery module, and/or a battery cell of the battery module; [0036] c. a capacity value representing an amount of energy that can be extracted from the battery module under a specified set of conditions (typically given in Amp-hours); [0037] d. one or more orientation values representing one or more orientations of the battery module, for example relative to gravity (e.g. x, y, z, pitch, yaw, and/or roll values); [0038] e. a thermal value; [0039] f. a battery module abuse condition, for example signaling a breach of the battery case and/or a tampering with the electronics and/or an acceleration of the battery module, e.g. above a specified threshold; [0040] g. an electrical current value, for example representing a current draw on the battery module, a battery cell series of the battery module, and/or a battery cell of the battery module; [0041] h. an energy usage value, for example representing a cumulative amount of energy extracted from the battery

module over a specified period of time (e.g. expressed in kilowatt hours (kWh)); [0042] i. a battery module theft condition, for example detected by physical breach of or damage to a security structure, such as a lock, upon or associated with the battery module, and/or detected by a usage pattern of the battery module that differs from an historical or expected usage pattern of the battery module, and/or detected by a location value for the battery module that differs from an historical or expected location value; [0043] j. a location value representing the location of the battery module, for example as detected using a global positioning satellite (GPS) system; [0044] k. an alarm value or history, e.g. representing a record of alarms or faults experienced by the battery module; [0045] l. a ganged/unganged value, e.g. representing a detection of whether or not the battery module is ganged together with one or more other battery modules in series or in parallel; [0046] m. a transportation usage condition providing an indication that the battery module is being and/or has been used to power a vehicle or other transportation device, for example as detected by a pattern of accelerations of the battery module, and/or a speed of the battery module, and/or a change in the location(s) of the battery module over time; [0047] n. a stationary usage condition providing an indication that the battery module is being and/or has been used in a stationary position, for example as detected by a lack of accelerations of the battery module during a period of time, and/or a lack of change in the location of the battery over time; [0048] o. a replace and/or recycle condition, e.g. as detected by an age of the battery module and/or an operative condition of the battery module.

[0049] As also discussed above, battery module(s) **202** are enabled for remote command and control by command center **236** through electromagnetic signaling received by battery module(s) **202**. In this regard, signaling to the battery module can in some forms be representative of one of, any combination of some of, or all of: [0050] signaling a user through display **224** of module **202**; [0051] signaling a user through lights **242, 244, 246**; [0052] managing the use and/or charging of battery cells **206**; [0053] disabling and/or enabling the draw of power from battery module **202** by a user (e.g. using those mechanisms therefor discussed herein).

[0054] System **200** can also include a user database **260** in which information correlated to a user or a plurality of users of battery modules **202** can be stored in memory, and from which such information can be retrieved by controller(s) **238**. User information can include a name or names associated with an account for the battery module(s), an account number, contact information (e.g. residence or business address(es), cellular or other telephone contact number(s); computer-based contact information such as email address(es), and/or historical data regarding the account such as payment history, current payment status, and others. Appendix A and Appendix B below refer to user information in several instances and it will be understood that in some embodiments such information can be stored in, and retrieved from, user database **260**. Such retrieval, and actions to the user (e.g. notifications) or to the battery module(s) **202** can be automatically implemented by controller(s) **238** (e.g. in response to a rule or algorithm) or in other embodiments can be implemented by an input command from a person through input **240**. In certain embodiments herein, the battery module(s) **202** can be disabled through a disable mechanism as described herein, such that a user can no longer draw power from the battery module(s) **202**. This can be in response to a payment condition of an account correlated to the user. For example, where use of the battery module(s) **202** is on a pre-paid basis by a user correlated to the battery module(s) **202** (e.g. pre-paid for a period of time of usage of battery module(s) **202** or for a total amount of power (e.g. measured in kilowatt hours) drawn from battery module(s)), the battery module(s) can be disabled through the disable mechanism upon or after exhaustion of the pre-paid usage amount. Data and calculations related to total usage of battery module(s) **202** (e.g. in time period and/or in total power drawn) and comparison of the same to the total usage for which prepayment has been made can be stored and conducted locally by the electronics of battery module(s), and/or such storage and calculations can be conducted at command center **236** using controller(s) **238** and an associated memory device (e.g. solid state or other memory storage mechanisms). Appropriate notifications or

warnings in advance of such disablement can be provided to the user, for example as specified in Appendix A and/or Appendix B. These notifications or warnings can be communicated at the battery module(s) and/or to alternate or secondary contacts for a user (e.g. stored in user database **260**). As well, in some embodiments, system **200** can detect a current use condition of the battery module(s) **202** under which it would be unsafe to disable the battery module(s), and the remote disablement of the battery module(s) can be delayed until such current use condition has been terminated. For example, where battery module(s) **202** are detected to be currently in motion by global positioning satellite (GPS) tracking, accelerometer readings, or otherwise (e.g. where the motion is indicative of use to electrically power a car, scooter or other vehicle), the disablement command can be delayed until the battery module(s) is/are detected to be no longer in motion. Additionally or alternatively, a current use condition that would delay remote disablement may also be detected through sensing an external connection to the battery module(s) **202** through the electronics of the battery module(s) or otherwise, for example an external connection to a vehicle or to hospital equipment powered by the battery module(s). In other embodiments, a user or users of battery module(s) can pay for use of the battery module(s) in modes other than pre-payment, for example by periodic payment, for example monthly, semi-annually or annually. In similar fashion to that discussed above for pre-paid use of battery module(s) **202**, a user payment history in respect of such periodic payments due can be used to determine that the battery module(s) **202** will be disabled. Again, appropriate advance warnings or notifications can be provided to the user(s) and/or safety-related delays in disablement of the battery module(s) can be implemented by system **200**.

[0055] With reference now to FIG. **1B**, shown is a schematic illustration of one illustrative remote communication implementation between a battery module(s) **202** and a command center **236** of system **200** described herein. As depicted, the communication can include a computer identification authentication between the center **236** and the module(s) **202**, a data request from the command center **236**, a data send by the module(s) **202**, a data receipt confirmation from the center **236** to module(s) **202**, and a display of the data on a display at the command center. FIG. **1B** also specifies various types of data that may be involved in such a data request and transfer. It will be understood that such data requests and receipts can be used in implementing the various actions specified in Appendix A and/or Appendix B below.

[0056] Still additional details of detection, command and control features that can be incorporated in embodiments of system **200** are disclosed in Appendix A and Appendix B below.

[0057] Referring now generally to FIGS. **1C** to **9** together, shown is one embodiment of a battery module **20** that can be used in aspects of the present invention. Battery module **20** includes an outer battery case **22**, which is desirably composed of an impact-resistant polymeric material such as glass-reinforced polypropylene, a polymeric/carbon fiber composite, or the like. Battery module **20** also includes a reinforcing divider element **24** that is partially received within the interior of battery case **22**. The reinforcing divider **24** is desirably a monolithic, thermally-conductive member that serves multiple purposes, including structurally reinforcing the battery case **22** to provide mechanical stability to module **20**, as well as defining cavities for housing battery cells, such as lithium ion battery pouch cells **26**. Reinforcing divider **24** also defines a plurality of thermal transfer element **80** that are positioned at least partially exterior of the interior enclosure of battery case **22**, as discussed further below. In the preferred embodiment shown, thermal transfer elements **80** extend beyond a bottom wall **42** defined by case **22**. Case **22** further defines a lip **23** extending below bottom wall **42** of case **22** and which is constructed and arranged to support the weight of module **20**. Lower lip **23** can extend below bottom wall **42** sufficiently to reduce or eliminate any weight to be supported by thermal transfer elements **80** when module **20** is supported on a surface by lip **23**. For these purposes, lower lip **23** can extend below bottom wall **42** a distance that is at least equal to, and desirably greater than, the distance that thermal transfer elements **80** extend below bottom wall **42**. Lip **23** can define at least one opening, and desirably a plurality of openings

25, when module **20** is resting on a surface. This can serve to allow ventilation of the space underneath module **20**, in which thermal transfer elements **80** reside, when module **20** is resting on a surface as supported by lip **23**.

[0058] Battery module **20** also includes a control board **28** or other means electrically connecting the battery cells **26** to provide cell group(s) in series and/or parallel, a positive terminal **30**, and a negative terminal **32** connected to the battery cell group(s). As well, battery module **20** can include a handle member **34** for carrying module **20**, which can as illustrated be comprised of flexible straps and a grip connected to the battery case **22**, for example by securing straps of handle **34** through openings **36** provided in a flange of case **22**. Also, positioned within the battery case **22**, between the case **22** and the reinforcing divider **24**, is shock dampening material **38**. Shock dampening material **38** can be provided as a singular unit or substance or can be provided at multiple, selected locations, interposed between and contacting surfaces of the reinforcing divider **24** and battery case **22**. In this manner a stable fit of the divider **24** within the case **22** can be provided, that will prevent shifting of the divider **24** within the case **22** and mechanically integrate the divider **24** and the case **22** so that the divider **24** can serve to reinforce the case **22** against impact or shock forces imparted to the exterior of case **22**. Desirably, reinforcing divider **24** will be reinforcingly mated in this fashion with at least the sidewalls and the bottom wall of the battery case **22**. Suitable shock dampening materials **38** may include, for example, polymer foam materials such as PORON® Urethane foam (Rogers Corporation, USA) and/or elastomeric and/or polymeric substances (e.g. silicone or other rubbers), which may also serve as adhesives to bond the reinforcing divider **24** to the battery case **22**.

[0059] Turning now to a more detailed discussion of various components of the battery module **20**, shown in FIG. **4** is a partial cut-away view of the battery module **20** illustrated in FIGS. **1** to **3**. Battery case **22** includes a lower portion or tub **22A** and an upper portion or cap member **22B**. Tub **22A** defines an internal volume generally larger than that of cap member **22B**, with tub **22A** including side walls **40** connected to bottom wall **42**. Bottom wall **42** defines a plurality of openings **42A** (see FIGS. **3** and **8**), such as slots, which are arranged to receive therethrough projecting thermal transfer members **80** defined by the reinforcing member **24**. Case tub **22A** also defines a peripheral flange **44**, which can be used for sealing engagement with battery case cap member **22B**. Peripheral flange **44** can define a cavity such as a slot **46** (shown in phantom in FIG. **4**; see also enlarged cutaway section in FIG. **4A**, and FIG. **8**), for receipt of a compressible seal member **48** (phantom, FIG. **4**) to aid in creating a sealed environment, desirably a hermetically or other moisture-proof sealed environment, within the assembled battery case **22**. Battery case cap member **22B** defines sidewalls **50** and a top wall **52**. Top wall **52** defines a number of openings including a first opening **54** for receipt of a display screen **64** for displaying information, for example about the status of one or more conditions of battery module **20**. Top wall **52** also defines openings **56** and **58** for receipt of and providing access to positive **30** and negative **32** terminals of battery module **20**. Cap member **22B** can also in certain embodiments define terminal covers **56A** and **58A** which are arranged to reversibly cap or cover openings **56** and **58**, respectively, to enable selective protection of terminals **30** and **32** from environmental conditions such as contaminants, moisture, or others. Covers **56A** and **58A** can for example be caps that are secured to capping member **22B** by a molded, living hinge. Cap member **22B** also defines a peripheral flange **60** arranged to mate with peripheral flange **44** of tub **22A**. Peripheral flange **60** can define a cavity **62** such as a slot (shown in phantom, FIG. **4**; see also enlarged cut away section in FIG. **4A**, and FIG. **9**), for alignment with slot **46** of flange **44** of tub **22A**, for capturing and compressing the seal member **48** within a seal cavity defined by cavities **46** and **62** combined. In the assembly of battery casing **22**, the tub **22A** and cap member **22B** can be attached to one another in any suitable manner including for example the use of an ultrasonic weld between flanges **44** and **60**, respectively, the use of adhesive or bonding agents, the use of suitable connectors such as bolts and nuts, or any other suitable means.

[0060] Within the battery case **22** when assembled, electronic control board **28** electrically connects pouch cells **26**, for example in series and/or parallel in groups. This electrical connection can be made in any known manner including for example by ultrasonic welding of the positive and negative electrodes of the pouch cells **26** to control board tabs, where the control board contains the necessary conductive materials to arrange the connected pouch cells **26** in series and/or parallel as desired. In addition to providing the cell connectivity, the control board **28** can contain circuitry, software and/or other components for sensing and/or communications, cellular or global positioning satellite signaling for purposes of locating or tracking module **20**, and/or for data transfer, including in some embodiments two-way data transfer. Some or all sensed conditions of the cell, or information relating to communications, signaling or data transfer, can be displayed on a display **64** coupled to control board **28**. Display **64** may, for example, be a liquid crystal display (LCD). Such a display **64** may, for example, display the state of charge and/or state of health of the battery module or of individual or groups of battery cells within the battery module **24**, and may for example common indicators of health of charge such as a selected color like red, yellow, or green to signify a state of charge or health of the battery module **20**. The inclusion of electronics for communication and data transfer can provide for the generation of electromagnetic signals by the module **20** or other modules of the invention for remote reporting and/or data collection with regard to sensed conditions of and/or location of the battery module **20** and potentially also for receiving signals for control of operation(s) of the module **20** from remote locations, and can for example utilize cellular and/or satellite communications networks for these purposes. These electronics are incorporated into the control board (e.g. control board **28**) in certain embodiments.

[0061] Control board **28** is also electrically connected to a positive electrical terminal **30** and a negative electrical terminal **32**, which as will be understood are ultimately connected to the pouch cell or cells **26**, and which provide electrical access to draw upon the electrical potential stored by battery module **20**. Terminals **30** and **32** can provide either male or female type connectors for connecting to cables or other conductive members, with female connectors being provided in some preferred embodiments. It will be understood that as located upon control board **28**, terminals **30** and **32** will be arranged to correspond in location with openings **56** and **58** in cap member **22B**, respectively, and that display **64** will be arranged to correspond in location with opening **54** in cap member **22B**. It will also be understood that these components **66**, **68** and **64** can be sealingly received in or through openings **56**, **58** and **54**, potentially with the use of gaskets or sealants or the like, to facilitate a sealed environment within case **22**.

[0062] With particular reference to FIGS. 5 to 7, shown are front, top and bottom views, respectively, of multi-purpose reinforcing divider **24**. Divider **24** includes a plurality of wall members **70** which in the depicted embodiment are arranged in generally parallel fashion relative to one another. Wall members **70** are connected by a set of transverse wall members **72** which retain wall members **70** in positions spaced from one another. Cross walls **72** can in certain embodiments be thicker than dividing wall members **70**. Reinforcing divider **24** thereby defines a plurality of cavities **74**, in the depicted embodiment in the form of slots, which are defined between sidewalls **76** of adjacent wall members **70**. A lower wall **78** of slots **74** is defined by an upper surface of the cross wall members **72**. Reinforcing divider **24** also defines a plurality of thermal transfer elements **80** which are connected to cross wall members **72**. In the depicted embodiment, thermal transfer elements **80** generally form continuing portions of wall members **70** occurring on the opposite side of cross walls **72**. Other arrangements are of course possible within the scope of the invention. Thermal transfer elements **80** include sidewalls **82** and a bottom wall surface **84**, to be exposed exterior of the interior enclosure defined by battery case **22** in the depicted embodiment (See e.g. FIG. 1, bottom). As shown particularly in FIG. 7, thermal transfer members **80** are provided in two rows. This arrangement can be provided by modification of an extruded member (e.g. extruded aluminum or other metal) extruded to a continuous shape having the profile shown in FIG. 5. This modification can include removal of sections of the lowermost wall portions from which thermal

transfer elements are formed (see dotted-line rectangular boxes in FIG. 7 representing areas where extruded wall sections have been removed), for example by machining, to result in the rows of elements **80** as shown. Appropriate finishing of the machined or otherwise modified extruded piece, for example using polishing or other smoothing operations, can be applied in the manufacture of the reinforcing divider **24**.

[0063] Reinforcing divider **24**, in the preferred embodiment depicted, also includes integrated external reinforcing scaffolding structures **86** and **88**. Reinforcing scaffolding structures **86** and **88** each include a plurality of elongate scaffold walls **90**, **92**, **94** extending in a first direction and a plurality of cross-scaffold walls **96** extending in a second direction transverse to the first direction, and connecting the elongate scaffold walls **90**, **92**, **94**. Reinforcing scaffolding structures **86** and **88** can thereby define a plurality of closed cells **98**. The outermost walls **94** of scaffolding structures **86** and **88** can extend non-parallel to walls **90** and **92** and to walls **70**, and in preferred embodiments are each inclined at an angle to position them, when divider **24** is received in battery case **22**, in a substantially parallel plane to adjacent sidewalls of the battery case **22** that the walls **94** will reinforce. Scaffolding structures **86** and **88** can provide enhanced mechanical strength and increase the resistance of the reinforcing divider **24** to damage from impact forces directed at the outer surfaces of scaffolding structures **86** and **88**.

[0064] Shown in FIGS. **8** and **9** are views illustrating the interior regions of the battery case tub **22A** and the battery case cap member **22B**, respectively. In FIG. **8**, the plurality of bottom wall openings **42A** can be clearly seen. As shown, the openings **42A** are provided in two sets extending along the length of tub **22A**, divided by a central wall region **100** which spans between first lateral end wall portion **102** and second lateral end wall portion **104**. As shown, rows of dimensionally identical openings, in the depicted embodiment slots, occur on either side of central wall portion **100**. It is preferred that the bottom wall of tub **22A**, as shown, includes at least one internal rib or wall portion **100** spanning between sets or groups of openings, to provide structural stability to the bottom wall of tub **22A**. As will be understood, the thermal transfer members **80** of reinforcing divider **24** are constructed and arranged to correspond in location to the openings **42A** in the bottom wall **42** of tub **22A**, and to be receivable through the openings **42A**. A layer or bead of shock dampening material (**38**, see previous Figs.) can be provided on the inner surface of bottom wall **42** of tub **22A**, to contact the lower surfaces of cross walls **72** and **96** of the reinforcing divider **24** when assembled into tub **22A**. This shock dampening material may also optionally be sufficient to create a sealed periphery around the **42A** collectively, in groups, or individually. This can provide a lower seal for the interior region of case **22** residing above and housing the battery cells **26** and potentially electronic components of control board **28**, such that gases or other potential contaminants passing into openings **42A** do not pass into such interior region. Alternatively, separate sealing agents or mechanisms can be used.

[0065] Also shown in FIG. **8** is a plurality of slotted retainers **102** for capturing the lateral edge portions **70A** (see FIG. **6**) of walls **70** of reinforcing divider **24**. Slotted retainers **102** each define a slot having a back wall **104** that is inclined relative to the inner surface **40A** of the adjacent sidewalls **40**, with back wall **104** preferably extending perpendicular to bottom wall **42** and thus situating in a substantially vertical plane when bottom wall **42** is positioned in a substantially horizontal plane. Further, the back walls **104** of retainers **102** aligned with one another on opposed sides of the tub **22A** (e.g. retainers **102A** and **102B** as labeled in FIG. **8**) are preferably spaced a distance from one another such that the lateral edge portions of walls **70** of retainer **24** are captured in the respective slots of the retainers, potentially frictionally engaging such lateral edge portions either with walls **104** and/or walls **106**, and/or with a shock dampening material (e.g. an adhesive and/or foam material as described herein) received within the slots of the retainers **102**. This further facilitates a mechanical integration of reinforcing divider **24** and battery case **22**. In the preferred battery module **20**, the slot back walls **106** and the lateral edges of reinforcing divider walls **70** both extend substantially vertically when module **20** is placed with the lower surface of the bottom lip

23 residing upon a horizontal surface.

[0066] Referring now particularly to FIG. **9**, located on the interior of cap member **22B** are a plurality of reinforcing struts or ribs that increase the structural integrity and strength of the cap member **22B**. In the depicted cap member **22B**, primary, thicker ribs **110** divide portions of the cap member **22B** into generally quadrilateral segments, and are interconnected by secondary, thinner ribs **112** which define a plurality of triangular enclosures. The ribs, e.g. **110** and **112**, can be molded integrally with the cap member **22B**, and can for example be about 1 to 3 mm in height as they extend inward from the inner surface of the outer wall of the case cap member **22B**. Corresponding or similar reinforcing rib structures are also desirably provided on the interior of battery case tub **22A**.

[0067] In operation, battery module **20** provides a highly protected, thermally managed condition for battery cells **26**. Thermal energy (heat) generated during operation of battery cells **26** can transfer to walls **70** of reinforcing divider and from there to thermal transfer elements **80** exposed externally of the sealed environment of the case **22** in which cells **26** are housed. Similarly, heat from the environment external of the battery can be transferred to battery cells **26** where cells **26** are cooler than the external environment, as may occur during charging or a non-operational state of cells **26**. In some embodiments, a thermally conductive material, for example a thermally conductive adhesive material, can be positioned between and potentially contact and/or adhere the outer surfaces of cells **26** and the walls **70**. The thermal management provided can aid in optimizing the performance of cells **26**. In the illustrated embodiment, the thermal management of cells **26** is passive thermal management, in that no active heating or cooling fluids are circulated through or against the thermally-conductive reinforcing divider **24**. It will be understood that active heating or cooling could be applied to reinforcing divider **24** in other embodiments, and divider **24**, e.g. walls **70** thereof, may optionally be modified with grooves, internal lumens or cavities, to circulate heating/cooling fluids in such other embodiments. Passive thermal management is preferred, as it simplifies the design and operation of battery module **22**.

[0068] It will also be understood that although the battery module **20** above has been illustrated and described in conjunction with prismatic lithium ion pouch cells **26**, other battery cells may also be used including cylindrical lithium ion battery cells (in which case reinforcing divider **24** may be modified to define cylindrical cavities for housing the cells), as well as battery cells employing other battery chemistries. Further, it will be understood that a number of lithium ion cell constructions and chemistries are known and available for use in embodiments of the invention, including both wet electrolyte and solid electrolyte (e.g. polymeric) designs.

[0069] Further, additional embodiments of the invention are provided wherein a reinforcing divider, potentially monolithic and thermally conductive as described herein, is received completely within a sealed interior of a battery case, and thus providing no externalized thermal transfer elements. Such reinforcing dividers can nonetheless facilitate robust, mechanically stable battery module constructions, particularly where the reinforcing divider reinforces the battery case with a shock dampening material therebetween. Still further, embodiments are provided where the battery module has the structure shown for module **20**, with externalized thermal transfer elements **80** beneficially extending below a bottom wall **42** of the battery case and even more beneficially being protected by a lower lip **23**, but wherein the reinforcing divider is not monolithic but rather can be made from multiple pieces connected together by screws, bolts or other connection mechanisms. Module **20** and modified versions thereof, with externalized transfer elements extending only out of a bottom wall, provide efficient thermal management while minimizing exposure of the thermal transfer elements **80** for potentially injurious contact with users of the module, or damaging contact with other structures.

[0070] Particularly beneficial battery modules of the invention will exhibit high energy densities (power-to-weight ratios), with energy densities of at least 50 watt-hours per kilogram (Wh/kg), and generally in the range of 70 Wh/kg to 200 Wh/kg, being contemplated in certain embodiments

herein. In preferred modes, the battery module will exhibit good portability, weighing less than about 12 kilograms, typically in the range of about 10 to about 12 kilograms. As well, desirable storage capacities for the battery module will be at least 1 kilowatt hour (kWh), typically in the range of about 1.5 kWh to about 2.5 kWh. A range of operational voltage capacities may be embodied, including for example at least about 36 volts direct current (DC), and typically about 48 to about 60 volts DC.

[0071] The uses of the terms “a” and “an” and “the” and similar references in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

[0072] While the invention has been illustrated and described in detail in the drawings and foregoing description, the same is to be considered as illustrative and not restrictive in character, it being understood that only the preferred embodiment has been shown and described and that all changes and modifications that come within the spirit of the invention are desired to be protected. In addition, all references cited herein are indicative of the level of skill in the art and are hereby incorporated by reference in their entirety.

Appendix A

User Information Account Processes

User Care and Billing

BTM (Battery Module)

TABLE-US-00002 Item Display Action Account Account number Identify account and align with historical data User Information User name, address, time Identify account with name and with energy company align with historical data BTM Serial All BTM serial numbers Align or match historical data Number(s) associated with account between BTM and user account number number and report BTM Technical Iconic models of BTMs Provide all historical data of Data associated with account BTMs upon selection Current Balance Account(s) monetary If greater than zero then send balance “green” signal for status. If equal to zero or less send “red” signal for status Payoff Balance Only if negative value Report negative balance Previous Bill Date and amount of Report previous bill with payment due date Last Payment Date and amount Report Next Billing Payment due date, days Report Cycle in between Pending Bill Date and amount Report Exists Pending Date and amount Report Payments Payment History 1 month, 3 month, 6 Upon selection criteria display month, 9 month, 12 data month, 2 year, 4 year, historical (options) User Payment 1, 2, 3, 4, or 5 (value) associate highest score/value with Score green color = 1; blue color = 2; yellow color = 3, gray color = 4, red color = 5 (poorest score/value) Action to User Display notice, action Action can vary from report (based on and corresponding generation, status text, to payment) customer classification terminating service (shutting off or disabling BTM through disable mechanism of BTM) BTM GPS Global / Regional / Local Provide dotted line of BTM Tracking Map Auto-Pay Active yes / no Appears when the account has an automatic payment option effective on the current date and appears if Auto-Payment functionality is enabled. Warnings series of boolean fault based on boolean series correlate codes (10 lights) to specific warning statements Interruptions database of interruptions Report or faults Power Off Sleep Mode Indication Sleep Mode

Appendix B

BTM Management and Control

Diagnostics, Data, Control

BTM (Battery Module)

TABLE-US-00003 Action to Reaction Action to External Action to Item Display HQ in RDAAC

BTM User Battery Value if less than X, analyze green, turn on Voltage or higher than and place green corresponding Y, then alert value in flashing, lighting on BTM “action” yellow, with text/digital step to HQ yellow display to describe and BTM flashing, condition (e.g., red, red “normal voltage”, flashing* “low voltage”, “voltage fault”, etc.) - if red flashing conditionally plan to turn BTM off in predetermined manner; optional or alternative notification to secondary contact for user stored in user database (e.g. cellular telephone, email) Cell Voltage Value if less than X, analyze or higher than and place Y, then alert value in “action” step to HQ Battery Temp Value if less than X, analyze green, turn on or higher than and place green corresponding Y, then alert value in flashing, lighting on BTM “action” yellow, with text/digital step to HQ yellow display to describe and BTM flashing, condition (e.g., red, red “normal temp flashing* “low/high temp”, “temp fault”, etc.) - if red flashing conditionally plan to turn BTM off in predetermined manner; optional or alternative notification to secondary contact for user stored in user database (e.g. cellular telephone, email) Cell Temp Value if less than X, analyze or higher than and place Y, then alert value in “action” step to HQ Battery Value(s) poll analyze green, turn on Capacity frequently, and place green corresponding and if less value in flashing, lighting on BTM than X, or “action” yellow, with text/digital higher than step to HQ yellow display to describe Y, then alert and BTM flashing, condition (e.g., red, red “normal capacity” flashing* “low capacity”, “capacity fault”, etc.) - if red flashing conditionally plan to turn BTM off in predetermined manner; optional or alternative notification to secondary contact for user stored in user database (e.g. cellular telephone, email) Position x, y, z, if less than X, analyze pitch, yaw, Y, Z, P, Y, R, and place roll or higher than value in X, Y, Z, P, Y, “action” R then alert step to HQ Thermal Value if less than analyze Management T1, or higher and place than T2, then value in alert “action” step to HQ Abuse Breach alert, shut analyze green, turn on (on/off) down and place green corresponding value in flashing, lighting on BTM “action” yellow, with text/digital step to HQ yellow display to describe and BTM flashing, condition (e.g., red, red “normal flashing* operation”, “slight abuse”, “abuse fault”, etc.) - if red flashing condition plan to turn BTM off in predetermined manner; optional or alternative notification to secondary contact for user stored in user database (e.g. cellular telephone, email) Current Value if less than X, analyze green, turn on or higher than and place green corresponding Y, then alert value in flashing, lighting on BTM “action” yellow, with text/digital step to HQ yellow display to describe and BTM flashing, condition (e.g., red, red “normal voltage”, flashing* “low voltage”, “voltage fault”, etc.) - if red flashing condition plan to turn BTM off in predetermined manner; optional or alternative notification to secondary contact for user stored in user database (e.g. cellular telephone, email) kWh Cumulative report hourly analyze green, turn on throughput value daily, weekly, and place green corresponding monthly, value in flashing, lighting on BTM annually “action” yellow, with text/digital step to HQ yellow display to describe and BTM - flashing, kWh throughput HQ to store red, red status, rate, all kWh flashing* payment status, data in condition (e.g., historical “normal voltage”, database “low voltage”, aligned “voltage fault”, etc.) - with user if red flashing data, etc. condition plan to turn BTM off in predetermined manner; optional or alternative notification to secondary contact for user stored in user database (e.g. cellular telephone, email) recycling physical Remaining analyze age, status time of 20 and place year life value in “action” step to HQ anti-theft Breach alert, shut analyze green, turn on (on/off) down and place green corresponding value in flashing, lighting on BTM “action” yellow, with text/digital step to HQ yellow display to describe and BTM flashing, condition (e.g., red, red “normal owner” flashing* “owner breach underway”, “owner fault”, etc.) - if red flashing condition plan to turn BTM off in predetermined manner; optional or alternative notification to secondary contact for user stored in user database (e.g. cellular telephone, email) Global location verify,

analyze Positioning coordinates confirm/alert and place Satellite (GPS) value in “action” step to HQ alarms record of store data analyze alarms with BTM and place faults, etc. and User(s) value in “action” step to HQ single BTM Value send analyze (non-ganged) indication of and place use value in “action” step to HQ ganged Value send analyze indication of and place use, and value in connected “action” items step to HQ transportation Value send analyze use of BTM indication of and place proper use value in and map of “action” daily travel step to HQ stationary use Value send analyze of BTM indication of and place proper use value in and map of “action” location step to HQ use strategy Value(s) projected analyze for BTM versus actual and place use profiles value in “action” step to HQ capacity Value(s) projected analyze strategy for versus actual and place BTM use profiles value in “action” step to HQ historical data Value(s) projected analyze and use for versus actual and place BTM use profiles value in “action” step to HQ energy Value(s) projected analyze management versus actual and place strategy for use profiles value in BTM “action” step to HQ external Protocols confirm analyze connection(s) & Match protocol and place to BTM connection - value in send yes or “action” no step to HQ *green = perfect condition green flashing = leaving prefect condition and moving toward caution yellow = caution yellow flashing = leaving caution status and moving toward alert or fault status red = alert/fault status red flashing = system shut down is imminent

Claims

1. A battery module for remote command, comprising: a battery case; a plurality of battery cells received within the battery case, a negative electrical terminal electrically coupled to the battery cells, a positive electrical terminal electrically coupled to the battery cells, the battery cells providing a voltage potential between the positive electrical terminal and the negative electrical terminal; electronics received within the battery case and configured to enable electromagnetic signaling to and from the battery module; and a disable mechanism responsive to the electromagnetic signaling to the battery module and operable to disable use of said voltage potential.
2. The battery module of claim 1, wherein the disable mechanism interrupts electrical continuity between the positive electrical terminal and the negative electrical terminal.
3. The battery module of claim 1, wherein the disable mechanism is operable to shield at least one of the positive electrical terminal and the negative electrical terminal from external electrical contact.
- 4-14. (canceled)
15. A transportation management system, comprising: a battery module according to claim 1; and an electric vehicle powered by the battery module; a controller remote from and communicatively coupled to the battery module and configured to receive the electromagnetic signaling from the battery module and transmit the electromagnetic signaling to the battery module; wherein the electromagnetic signaling from the battery module includes electromagnetic signaling representative of a current use condition of the battery module, and wherein the current use condition includes a condition that the battery module is in motion.
16. The system of claim 15, wherein: the controller is programmed to delay electromagnetic signaling to the battery module to operate the disable mechanism to disable use of the voltage potential in response to receiving electromagnetic signaling from the battery module representing the condition that the battery module is in motion.
17. The system of claim 15, wherein: the controller is programmed to delay electromagnetic signaling to the battery module to operate the disable mechanism in response to receiving electromagnetic signaling from the battery module representing a condition that the battery module is electrically connected to an external device.
18. The system of claim 15, wherein: the system includes electronic memory including account

information of a user, the account information including a user contact; and the controller is programmed to access the account information and send advance notice to the user contact of an upcoming operation of the disable mechanism to disable use of the voltage potential.

19. The system of claim 18, wherein the user contact is an email contact or cellular telephone contact.

20. The system of claim 15, wherein the electric vehicle is an electric scooter.

21. A vehicle, comprising: a battery module according to claim 1; and an electric vehicle powered by the battery module.

22. The vehicle of claim 21, wherein: the battery module is communicatively coupled to a remote controller, wherein the electromagnetic signaling to the battery module is sent by the remote controller and the electromagnetic signaling from the battery module is received by the remote controller; the electromagnetic signaling from the battery module includes electromagnetic signaling representative of a current use condition of the battery module, and wherein the current use condition includes a condition that the battery module is in motion and a condition that the battery module is stationary.

23. The vehicle of claim 22, wherein: the remote controller is programmed to delay electromagnetic signaling to the battery module to operate the disable mechanism in response to (i) receiving electromagnetic signaling from the battery module representing the condition that the battery module is stationary and/or (ii) representing a condition that the battery module is electrically connected to an external device.

24. The vehicle of claim 22, wherein the electric vehicle is an electric scooter.

25. A management system, comprising: a battery module configured for connection to an electric vehicle to power the electric vehicle, the battery module including a battery case, a plurality of battery cells received within the battery case, a negative electrical terminal electrically coupled to the battery cells, a positive electrical terminal electrically coupled to the battery cells, and electronics configured to enable electromagnetic signaling to and from the battery module; a disable mechanism responsive to the electromagnetic signaling to the battery module to disable power draw from the battery module; a controller remote from and communicatively coupled to the battery module and configured to receive the electromagnetic signaling from the battery module and transmit the electromagnetic signaling to the battery module; and wherein the electromagnetic signaling from the battery module includes electromagnetic signaling representative of a current use condition of the battery module, and wherein the current use condition is that the battery module is in motion.

26. The system of claim 25, wherein: the controller is programmed to delay electromagnetic signaling to the battery module to operate the disable mechanism in response to receiving electromagnetic signaling from the battery module representing the condition that the battery module is in motion.

27. The system of claim 25, wherein: the controller is programmed to delay electromagnetic signaling to the battery module to operate the disable mechanism to disable use of the voltage potential in response to receiving electromagnetic signaling from the battery module representing a condition that the battery module is electrically connected to an external device.

28. The system of claim 25, wherein: the system includes electronic memory including account information of a user, the account information including a user contact; and the controller is programmed to access the account information and send advance notice to the user contact of an upcoming operation of the disable mechanism to disable use of the voltage potential.

29. The system of claim 28, wherein the user contact is an email contact or cellular telephone contact.

30. The system of claim 25, also including the electric vehicle.

31. The system of claim 30, wherein the electric vehicle is an electric scooter.
